LineaLinkR

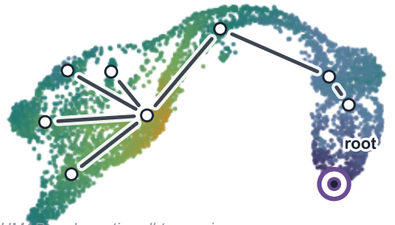
Trajectory-resolved temporal ligand–receptor communication with bifurcation-aware roles

single-cell snapshot → signaling along reconstructed trajectories

01 · INPUT 1

Snapshot state space

counts → PCA / kNN graph



UMAP · schematic cell-type spine

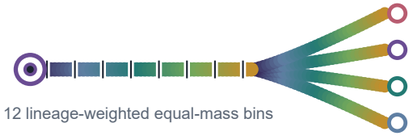
early — late

counts + LR prior

02 · TRAJECTORY 2

Lineages & time

geodesic pseudotime → bins



12 lineage-weighted equal-mass bins

- expression-only endpoint-selected attractor root
- Dijkstra geodesic pseudotime
- MST lineages + soft weights → ψ_{il}

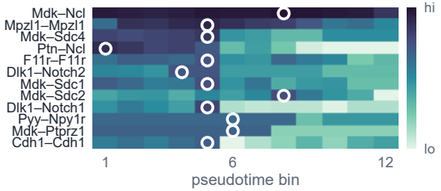
4 lineages recovered

03 · SCORING 3

Temporal LR programs

peak-lineage envelope · display gaps filled

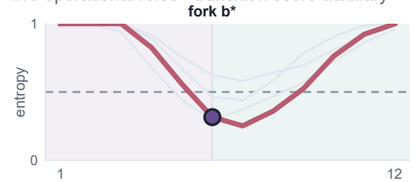
$$A_{LR} = \frac{\sqrt{L \cdot R}}{S + \sqrt{L \cdot R}}$$



04 · INTERPRET 4

Bifurcation → roles

five operational roles · transition score auxiliary



pre-fork

- Instructive
- Lateral

post-fork

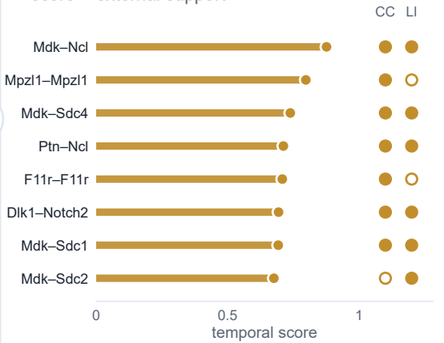
- Reinforcing
- Feedback

- Bystander

05 · EVIDENCE 5

Ranked programs

score + external support



EVIDENCE real and simulated data · blind ordering · external biological checks

37 broad evaluation	**22** common-data accuracy	**16** branch recovery	**10** method-specific adapters	**5 · 12** timing taxonomy
registered dataset slots	**ordering sets**	**branching GT**	**eligible TI comparators**	**operational roles · bins**